
NOTES FOR RENORMALIZATION FLOW

Based on the paper by A.Dunlap and Cole

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Contents

1 Setup

1.1 Checklist

1. Theorem 3.2.2. I should be some interval such that $\sigma > 0$.
2. Finish the argument that Sym is a vector space.
3. Well-definedness of the Wiener integral.

1.2 Semilinear SHE

We consider the semilinear stochastic heat equation

$$du_t^\rho = \frac{1}{2}\Delta u_t^\rho dt + \gamma_\rho \sigma(u_t^\rho) dW_t^\rho, \quad t > 0, x \in \mathbb{R}^2$$

Here σ is a Lipschitz nonlinearity and $dW_t^\rho(x)$ is a Gaussian noise that is white in time and correlated in space at scale $\rho^{1/2} \ll 1$. We are interested in the pointwise behavior of $u_t^\rho(x)$ as $\rho \rightarrow 0$, which calls for an attenuation factor $\gamma_\rho \sim |\ln \rho|^{-1/2}$ due to critical scaling in two dimensions. In fact, we devote most of our attention to a variation on (??) in which we first multiply σ and then smooth the noise:

$$dv_t^\rho = \frac{1}{2}\Delta v_t^\rho dt + \gamma_\rho \mathcal{G}_\rho[\sigma(v_t^\rho) dW_t]$$

Definition 1.2.1.

(Space-time White Noise)

Let $dW = (dW_t(x))_{t \in \mathbb{R}, x \in \mathbb{R}^2}$ be a standard $\mathbb{R}^m$ -valued space-time white noise generating a temporal filtration $\{\mathcal{F}_t\}_{t \in \mathbb{R}}$. Writing $dW = (dW^1, \dots, dW^m)$ in components, then

$$\mathbb{E}[dW_t^i(x) dW_{t'}^{i'}(x')] = \delta_{i,i'} \delta(t - t') \delta(x - x')$$

Proposition 1.2.1. Construct a space-time white noise.

Definition 1.2.2.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and fix a target dimension $m \in \mathbb{N}$. The solution $v^\rho : \Omega \times \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^m$ is a random vector-valued function parametrized by the correlation parameter $\rho > 0$. We suppress the dependence of v^ρ on $\omega \in \Omega$.

Since v is vector-valued, our nonlinearity $\sigma : \mathbb{R}^m \rightarrow \mathbb{R}^{m \times m}$ is matrix-valued. Let $\mathcal{H}_+^m$ denote the set of nonnegative-definite symmetric real $m \times m$ matrices, equipped with the metric induced by the Frobenius norm

$$|A|_F^2 := \text{tr}(AA^T) = \text{tr}(A^2)$$

Let σ belong to the space $\text{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$.

Proposition 1.2.2. Show that $|\cdot|_F$ is a norm on $\mathcal{H}_+^m$.

Definition 1.2.3. Given $\tau \geq 0$, we define the heat operator

$$\mathcal{G}_\tau v = G_\tau * v$$

where $G_\tau = (2\pi\tau)^{-1} \exp(-\frac{|x|^2}{2\tau})$ denotes the standard heat kernel. Define the spatially-smoothed noise $dW_t^\rho = G_\rho * dW_t$.

Proposition 1.2.3. We have

$$\mathbb{E}[dW_t^{\rho,i}(x)dW_{t'}^{\rho,i'}(x')] = \delta_{i,i'}\delta(t-t')G_{2\rho}(x-x')$$

Proof. □

Definiton 1.2.4. Define

$$L(\tau) = \ln(1 + \tau) \quad \text{for } \tau \geq 0$$

and set

$$\gamma_\rho = \sqrt{\frac{4\pi}{L(1/\rho)}}$$

Definiton 1.2.5.

(Mild Solution 1)

A mild solution for (??) is a predictable random field v^ρ such that for all $s < t$, we have

$$v_t^\rho(x) = \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho \int_s^t \mathcal{G}_{t+\rho-r}[\sigma(v_r^\rho)dW_r](x)$$

which means

$$\begin{aligned} v_t^\rho(x) &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho \int_s^t \int G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y)dy \\ &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho \int_s^t \int G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y)dy \\ &= \mathcal{G}_{t-s}v_s^\rho(x) + \gamma_\rho \int \left(\int_s^t G_{t+\rho-r}(y)\sigma(v_r^\rho)(x-y)dW_r(x-y) \right) dy \end{aligned}$$

which can be interpreted as an Ito integral. We only look for the solution v_t^ρ in the spaces $\mathcal{X}_t^l$ of $\mathbb{R}^m$ -valued random fields z on $\mathbb{R}^2$ that are $\mathcal{F}_t$ -measurable and

$$\|z\|_l := \sup_{x \in \mathbb{R}^2} (\mathbb{E}|z(x)|^l)^{1/l} < \infty$$

Proposition 1.2.4. For any $l \geq 2$, there is a family of random operators $(\mathcal{V}_{s,t}^{\sigma,\rho})_{s < t}$ such that if $v_s \in \mathcal{X}_s^l$, then $v_t^\rho = \mathcal{V}_{s,t}^{\sigma,\rho}v_s$ is a mild solution of (??) for $t \geq s$. We often write $\mathcal{V}_t^{\sigma,\rho} := \mathcal{V}_{0,t}^{\sigma,\rho}$.

Shown by some standard fixed-point arguments.

Definiton 1.2.6. (Forward-backward SDE)

The system of SDE:

$$\begin{aligned} d\Gamma_{a,Q}^\sigma(q) &= J_\sigma(Q-q, \Gamma_{a,Q}^\sigma(q))dB(q), & a \in \mathbb{R}^m, 0 < q < Q \\ \Gamma_{a,Q}^\sigma(0) &= a, & a \in \mathbb{R}^m, Q \geq 0 \\ J_\sigma(q, b) &= [\mathbb{E}\sigma^2(\Gamma_{a,Q}^\sigma(q))]^{1/2}, & q \geq 0, b \in \mathbb{R}^m \end{aligned}$$

for B a standard $\mathbb{R}^m$ -valued Brownian motion and $A^{1/2}$ is the unique positive-definite matrix

square root of $A \in \mathcal{H}_+^m$.

Definiton 1.2.7. Given $\sigma \in \text{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$, let $\overline{Q}_{\text{FBSDE}}(\sigma) \in [0, \infty]$ be the supremum of all $Q \geq 0$ such that there is a continuous function $J_\sigma : [0, Q] \times \mathbb{R}^m \rightarrow \mathcal{H}_+^m$ satisfying the FBSDE and

$$\sup_{q \in [0, \overline{Q}]} \text{Lip}(J_\sigma(q, \cdot)) < \infty$$

and define for $M, \beta \in (0, \infty)$

$$\Sigma(M, \beta) := \{\sigma \in \text{Lip}(\mathbb{R}^m, \mathcal{H}_+^m) : |\sigma(u)|_F^2 \leq M + \beta^2 |u|^2 \text{ for all } u \in \mathbb{R}^m\}$$

Definiton 1.2.8.

(L^2 -subcritical)

A nonlinearity σ is L^2 -subcritical if $\sigma \in \Sigma(M, \beta)$ for some $M \in (0, \infty)$ and $\beta \in (0, 1)$ and $\overline{Q}_{\text{FBSDE}}(\sigma) > 1$.

1.3 Main Result

Theorem 1.3.1. Let σ be L^2 -subcritical. Fix $Q \in (0, 1)$ and define $\tilde{\sigma} := (1 - Q)^{1/2} J_\sigma(Q, \cdot)$ and $\tilde{\rho} := \rho^{1-Q}$. Then there is a new white noise $d\tilde{W}$ such that $\mathcal{G}_{\tilde{\rho}v}$ is an approximate mild solution of (??) with $(\tilde{\rho}, \tilde{\sigma}, d\tilde{W})$ in place of (ρ, σ, dW) .

Definiton 1.3.1. Denote $\mathcal{W}_2$ the Wasserstein-2 metric: for any two probability distributions μ, ν

$$\mathcal{W}_2(\mu, \nu) = \inf_{\pi} \left(\int |x - y|^2 \pi(dx, dy) \right)$$

where π has marginal distributions μ and ν .

Theorem 1.3.2. For each L^2 -subcritical σ and $\overline{T} \in [1, \infty)$, there is a constant $C(\sigma, \overline{T}) \in (e, \infty)$ such that for all $v_0 \in L^\infty(\mathbb{R}^2, \mathbb{R}^m)$, $t \in [\overline{T}^{-1}, \overline{T}]$, and $\rho \in (0, C^1)$, the solution v^ρ of (??) satisfies

$$\mathcal{W}_2(v_t^{\rho(x)}, \Gamma_{a,1}^\sigma(1)) \leq C \langle \|v_0\|_{L^\infty} \rangle \sqrt{\frac{\ln |\ln \rho|}{|\ln \rho|}}$$

where $(\Gamma_{a,1}^\sigma(q))_{q \in [0,1]}$ solves the FBSDE with $Q = 1$ and $a = \mathcal{G}_t v_0(x)$.

2 Well-posedness of the FBSDE

2.1 Main Goal

The main result is

Theorem 2.1.1. Let $\sigma \in \text{Lip}(\mathbb{R}^m, \mathcal{H}_+^m)$. For any $Q \in [0, \overline{Q}_{\text{FBSDE}}(\sigma))$, we have

$$\overline{Q}_{\text{FBSDE}}(\sigma) \geq Q + \text{Lip}(J_\sigma(Q, \cdot))^{-2}$$

Definiton 2.1.1. $\mathcal{X}$ is the Banach space of $\mathcal{H}_+^m$ -valued continuous functions on $\mathbb{R}^m$ with the norm (and this norm is finite)

$$\|\sigma\|_{\mathcal{X}} := \sup_{x \in \mathbb{R}^m} \frac{|\sigma(x)|_F}{\langle x \rangle}$$

Denote $\langle a \rangle := (|a|^2 + 1)^{1/2}$ the Japanese bracket.

Proposition 2.1.2. Prove $\mathcal{X}$ is a Banach space.

Proposition 2.1.3. $\text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m) \subset \mathcal{X}$.

2.2 SDE Solution Theory

Definiton 2.2.1. Given a $\mathbb{R}^m$ -valued Brownian motion $(B(q))_{q \geq 0}$ adapted to a filtration $\{\mathcal{G}_q\}_{q \geq 0}$. For an adapted process Y on $[0, Q]$, a function $g : [0, Q] \times \mathbb{R}^m \rightarrow \mathcal{H}_+^m$ and a constant $a \in \mathbb{R}^m$, we define a new adapted process $\mathcal{R}_{a,Q}^g Y$ on $[0, Q]$ by

$$\mathcal{R}_{a,Q}^g Y(q) := a + \int_0^q g(Q - p, Y(p)) dB(p)$$

whenever this stochastic integral is defined. For $Q > 0$, define

$$\mathcal{A}_Q := \left\{ J : [0, Q] \times \mathbb{R}^m \rightarrow \mathcal{H}_+^m \text{ continuous} : \sup_{q \in [0, Q]} \text{Lip}(J(q, \cdot)) < \infty \right\}$$

Proposition 2.2.1. Fix $L < \infty$ and $Q \in (0, \infty)$ and suppose that $g \in \mathcal{A}_Q$ satisfies

$$\sup_{q \in [0, Q]} \text{Lip}(g(q, \cdot)) \leq L$$

Then, for any $a \in \mathbb{R}^m$, there is a unique strong solution $\Theta_{a,Q}^g$ to the SDE

$$\begin{aligned} d\Theta_{a,Q}^g(q) &= g(Q - q, \Theta_{a,Q}^g(q)) dB(q), \quad q \in [0, Q] \\ \Theta_{a,Q}^g(0) &= a \end{aligned}$$

The solution $\Theta_{a,Q}^g$ satisfies the moment bound

$$\sup_{q \in [0, Q]} \mathbb{E} |\Theta_{a,Q}^g(q)|^l < \infty \quad \text{for all } l \in [1, \infty)$$

Moreover, there exists a constant $C = C(L, Q)$ such that for any $Q' \in [0, Q]$ and any adapted

process Γ on $[0, Q']$, we have

$$\sup_{q \in [0, Q']} \mathbb{E} |\Gamma(q) - \Theta_{a, Q}^g(q)|^2 \leq C \sup_{[0, Q']} \mathbb{E} |\Gamma(q) - \mathcal{R}_{a, Q}^g \Gamma(q)|^2$$

and

$$\mathbb{E} |\Theta_{a, Q}^g(q) - \Theta_{\tilde{a}, Q}^g(q)|^2 \leq C |a - \tilde{a}|^2$$

Proof. Define $\mathcal{K}_{L, Q'}$ on adapted processes on $[0, Q']$ by

$$\sup_{q \in [0, Q']} e^{-L^2 q} (\mathbb{E} |\Gamma(q)|^2)^{1/2},$$

which makes $\mathcal{R}_{a, Q}^g$ a contraction. $\square$

2.3 Local Solution

Definiton 2.3.1. Let $Q > 0$ and let $\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m)$. We say that $J \in \mathcal{A}_Q$ is a *root decoupling function* for FBSDE on $[0, Q]$ if, for all $q \in [0, Q]$ and all $a \in \mathbb{R}^m$, we have

$$J(q, a) = [\mathbb{E} \sigma(\Theta_{a, q}^J(q))]^{1/2}$$

In this case, we also call J^2 the decoupling function. In the equation above, $\Theta_{a, q}^J$ is as in (??).

Definiton 2.3.2. For $\lambda \in (0, \infty)$, we define the set of functions

$$\Lambda(\lambda) := \{\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m) : \text{Lip}(\sigma) \leq \lambda\}$$

Proposition 2.3.1. Suppose that $\lambda \in (0, \infty)$, $\sigma \in \Lambda(\lambda)$, $Q_0 > 0$ and that J is a root decoupling function for FBSDE on $[0, Q_0]$. Then for all $Q \in [0, Q_0 \wedge \lambda^{-2}]$, we have

$$\text{Lip}(J(Q, \cdot)) \leq (\lambda^{-2} - Q)^{-1/2}.$$

Proof. $\square$

Lemma 2.3.2. Suppose that $c \in (0, \infty)$, $\overline{Q} < c^{-2}$ and $f : [0, \overline{Q}] \rightarrow [c^2, \infty)$ satisfies

$$f(Q) \leq c^2 \exp \left\{ \int_0^Q f(q) dq \right\}$$

for all $Q \in [0, \overline{Q}]$. Then

$$f(Q) \leq (c^{-2} - Q)^{-1} \quad \text{for all } Q \in [0, \overline{Q}]$$

Proof. Define $g(Q) = \int_0^Q f(q) dq$ and we have

$$g'(q) = f(q),$$

then

$$(1 - e^{-g(q)})' = g'(q) e^{-g(q)} = f(q) e^{-g(q)}$$

and hence

$$1 - e^{-g(Q)} = \int_0^Q f(q)e^{-g(q)} dq \leq c^2 Q$$

Therefore, we have

$$f(Q) \leq c^2 e^{g(Q)} \leq \frac{c^2}{1 - c^2 Q}$$

Here is another proof by (??), consider h defined on $[0, \overline{Q}]$ non-negative defined by

$$h(t) = \ln(f(t)/c^2),$$

then we have

$$h(t) \leq \int_0^t c^2 \exp(h(s)) ds.$$

Define

$$G(x) = -e^{-x} + C$$

and we have

$$h(q) \leq -\ln(1 - c^2 q)$$

for any $q \in [0, \overline{Q}]$, which means

$$f(Q) \leq c^2 / (1 - c^2 Q) = (c^{-2} - Q)^{-1}.$$

□

Definiton 2.3.3. Define

$$\mathcal{Q}_\sigma g(Q, a) = [\mathbb{E} \sigma^2(\Theta_{a,Q}^g(Q))]^{1/2},$$

where $(\Theta_{a,Q}^g(q))_{q \in [0, Q]}$ from (??). We note that a fixed point of Q_σ is a root decoupling function for (??). We also note that

$$|\mathcal{Q}_\sigma g(Q, a)|_F^2 = |\mathbb{E} \sigma^2(\Theta_{a,Q}^g(Q))|_F^2 = \mathbb{E} \text{tr} \sigma^2(\Theta_{a,Q}^g(Q)) = \mathbb{E} |\sigma(\Theta_{a,Q}^g(Q))|_F^2$$

Define the set of functions

$$\Lambda(M, \lambda) := \{\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m) : \text{Lip}(\sigma) \leq \lambda \text{ and } |\sigma(u)|_F^2 \leq M + \lambda^2 |u|^2 \text{ for all } u \in \mathbb{R}^m\}$$

For $Q_0 < \lambda^{-2}$, define the set of functions $\mathcal{Z}_{Q_0, M, \lambda}$ by

$$\begin{aligned} \mathcal{Z}_{Q_0, M, \lambda} = \{ & g : [0, Q_0] \times \mathbb{R}^m \rightarrow \mathcal{H}_+^m \text{ continuous,} \\ & g(q, \cdot) \in \Lambda((1 - \lambda^2 q)^{-2} M, (\lambda^{-2} - q)^{-1/2}) \text{ for all } q \in [0, Q_0] \} \end{aligned}$$

We will construct the root decoupling function J_σ as a fixed point of Q_σ in a certain $\mathcal{Z}_{Q_0, M, \lambda}$.

Proposition 2.3.3. Fix $\lambda, M \in (0, \infty)$ and $Q_0 \in [0, \lambda^{-2})$. For any $\sigma \in \Lambda(M, \lambda)$, there is a unique root decoupling function $J_\sigma \in \mathcal{A}_{Q_0}$. In particular, we have

$$\overline{Q}_{\text{FBSDE}}(\sigma) \geq \text{Lip}(\sigma)^{-2}.$$

Moreover, there is a $C = C(Q_0, M, \lambda) < \infty$ such that for any $g \in \mathcal{Z}_{Q_0, M, \lambda}$, we have

$$\sup_{q \in [0, Q_0]} \|(g - J_\sigma)(q, \cdot)\|_{\mathcal{X}} \leq C \sup_{q \in [0, Q_0]} \|(g - \mathcal{Q}_\sigma g)(q, \cdot)\|_{\mathcal{X}}$$

and indeed

$$\lim_{n \rightarrow \infty} \sup_{q \in [0, Q_0]} \|(\mathcal{Q}_\sigma^n g - J_\sigma)(q, \cdot)\|_{\mathcal{X}} = 0$$

where $\mathcal{Q}_\sigma^n$ denotes the n -fold iterated application of $\mathcal{Q}_\sigma$.

Proof. We know any decoupling function $J \in \mathcal{A}_{Q_0}$ has to be in $\mathcal{Z}_{Q_0, M, \lambda}$ for some $M', \lambda \in (0, \infty)$ $\square$

2.4 Extension of the Solution

Lemma 2.4.1. Let $\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m)$, whenever $0 \leq Q_1 \leq Q_2 < \overline{Q}_{\text{FBSDE}}(\sigma)$, we have for any $b \in \mathbb{R}^m$,

$$J_\sigma^2(Q_2, b) = \mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b, Q_2}^\sigma(Q_2 - Q_1))]$$

Proof. Remain. $\square$

Lemma 2.4.2. Let $\sigma \in \text{Lip}(\mathbb{R}^m; \mathcal{H}_+^m)$, whenever $0 \leq Q_1 \leq Q_2 < \overline{Q}_{\text{FBSDE}}(\sigma)$ and if

$$Q_2 - Q_1 < \overline{Q}_{\text{FBSDE}}(J_\sigma(Q_1, \cdot)),$$

we have

$$J_\sigma(Q_2, b) = J_{J_\sigma(Q_1, \cdot)}(Q_2 - Q_1, b) \quad \text{for all } b \in \mathbb{R}^m.$$

Proof. We have by (??) that, for any $Q \in [Q_1, \overline{Q}_{\text{FBSDE}}(\sigma))$ and any $b \in \mathbb{R}^m$,

$$J_\sigma^2(Q, b) = \mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b, Q}^\sigma(Q - Q_1))]$$

By assumption, there is a unique solution to the following FBSDE problem for $Q \in [0, Q_2 - Q_1]$:

$$\begin{aligned} d\Gamma_{b, Q}^{J_\sigma(Q_1, \cdot)}(q) &= J_{J_\sigma(Q_1, \cdot)}(Q - q, \Gamma_{b, Q}^{J_\sigma(Q_1, \cdot)}(q))dB(q), \quad q \in (0, Q) \\ \Gamma_{b, Q}^{J_\sigma(Q_1, \cdot)}(0) &= b \\ J_{J_\sigma(Q_1, \cdot)}(Q, b) &= (\mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b, Q}^{J_\sigma(Q_1, \cdot)}(Q))])^{1/2} \end{aligned}$$

Also, if given $Q \in [0, \overline{Q}_{\text{FBSDE}}(\sigma) - Q_1)$, (??) and (??) with Q replaced by $Q_1 + Q$ will yield

$$\begin{aligned} d\Gamma_{b, Q+Q_1}^\sigma(q) &= J_\sigma(Q + Q_1 - q, \Gamma_{b, Q+Q_1}^\sigma(q))dB(q), \quad q \in (0, Q) \\ \Gamma_{b, Q+Q_1}^\sigma(0) &= b \\ J_\sigma(Q + Q_1, b) &= (\mathbb{E}[\sigma^2(\Gamma_{b, Q+Q_1}^\sigma(Q))])^{1/2} = (\mathbb{E}[J_\sigma^2(Q_1, \Gamma_{b, Q+Q_1}^\sigma(Q))])^{1/2} \end{aligned}$$

which means $(\Gamma_{b, Q+Q_1}^\sigma(q)$ and $J_\sigma(Q + Q_1, b)$) will solve the previous FBSDE system, and by the uniqueness of FBSDE problem, we have

$$J_{J_\sigma(Q_1, \cdot)}(Q, b) = J_\sigma(Q + Q_1, b) \quad \text{for all } b \in \mathbb{R}^m \text{ and } Q \in [0, Q_2 - Q_1]$$

□

Proposition 2.4.3. For any $Q' \in [0, \overline{Q}_{\text{FBSDE}}(\sigma))$, we have

$$\overline{Q}_{\text{FBSDE}}(\sigma) \geq Q' + \overline{Q}_{\text{FBSDE}}(J_\sigma(Q', \cdot))$$

Proof. For $Q' < \overline{Q}_{\text{FBSDE}}\sigma$, there is a unique root decoupling function $J_\sigma \in \mathcal{A}_{Q'}$ for (??) on $[0, Q']$. Let $P \in [0, \overline{Q}_{\text{FBSDE}}(J_\sigma(Q', \cdot))]$, there is a unique RDF $J_{J_\sigma(Q', \cdot)} \in \mathcal{A}_P$ in (??) with σ replaced by $J_\sigma(Q', \cdot)$.

We wish to extend the function J_σ to the time interval by outting

□

3 Parabolic Equation for the decoupling function

3.1 Decoupling Flow

Definiton 3.1.1 (Parabolic Equation).

The following parabolic equation

$$\begin{aligned}\partial_q H(q, b) &= \frac{1}{2} [H(q, b) : \nabla_b^2] H(q, b) \\ H(0, b) &= \sigma^2(b)\end{aligned}$$

here for A, B matrices, we denote

$$A : B = \text{tr}[AB]$$

and there is an explicit formula

$$([H(q, b) : \nabla_b^2] H(q, b))_{ij} = \text{tr}[H(q, b) \nabla_b^2] H_{ij}(q, b) = \sum_{k, l=1}^m H_{kl}(q, b) \frac{\partial^2 H_{ij}}{\partial b_k \partial b_l}(q, b)$$

Proof. We know

$$\text{tr}[H(q, b) \nabla_b^2] = \sum_{k=1}^n \sum_{l=1}^n H_{kl}(q, b) \frac{\partial^2}{\partial b_k \partial b_l}$$

□

3.2 Main Theorem

Remark. The space

$$L^1(\mathcal{K}; W^{2, \infty}(U; \mathcal{H}_+^m))$$

means some functions which are L^1 with fixed space point and time varying and $W^{2, \infty}(U : \mathcal{H}_+^m)$ with fixed time point and space varying, with the norm

$$\|f\| = \int \|f(t, \cdot)\|_{W^{2, \infty}} dt.$$

For two normed spaces, their intersection will be equipped with a norm obtained by adding both the norms together.

Definiton 3.2.1. An *almost classical solution* to (??) on a time interval $[0, Q_0]$ is a continuous function $H : [0, Q_0] \times \mathbb{R}^m \rightarrow \mathcal{H}_+^m$ such that the following conditions hold:

1. For every compact $\mathcal{K} \subset (0, Q_0)$ and bounded open $U \subset \mathbb{R}^m$,

$$H_{\mathcal{K} \times U} \in L^1(\mathcal{K}; W^{2, \infty}(U; \mathcal{H}_+^m)) \cap C^1(\mathcal{K} \times U; \mathcal{H}_+^m).$$

where $W^{2, \infty}(U; \mathcal{H}_+^m)$ is the Sobolev space of functions on U taking values in $\mathcal{H}_+^m$ with weak second derivative measurable and essentially bounded.

2. We have $H(0, b) = \sigma^2(b)$ for all $b \in \mathbb{R}^m$.

3. We have

$$\partial_q H(q, b) = \frac{1}{2}[H(q, b) : \nabla_b^2]H(q, b) \quad \text{for almost all } (q, b) \in (0, Q_0) \times \mathbb{R}^m$$

Proposition 3.2.1. Fix $Q_0 \in (0, \overline{Q}_{\text{FBSDE}}(\sigma))$. Suppose H is an almost classical solution of (??) on $[0, Q_0]$ such that $\sup_{q \in [0, Q_0]} \text{Lip} \sqrt{H(q, \cdot)}$. Also assume that for each compact $\mathcal{K} \subset (0, Q_0)$, there exists a constant $C(\mathcal{K}) < \infty$ such that for all $R > 0$,

$$\|H|_{\mathcal{K} \times U_R}\|_{L^1(\mathcal{K}; W^{2, \infty}(U; \mathcal{H}_+^m)) \cap C^1(\mathcal{K} \times U; \mathcal{H}_+^m)} \leq CR^C.$$

Then $\sqrt{H} = J_\sigma$ on $[0, Q_0] \times \mathbb{R}^m$.

Theorem 3.2.2. Suppose $\sigma \in \text{Lip}(\mathbb{R}; \mathbb{R}^+)$ and $I := \{b \in \mathbb{R} : \sigma(b) > 0\}$ is an open interval. Suppose also that there exists $\beta \in (0, \infty)$ such that the applicable following condition holds.

- (i) If I is a bounded interval, then there is a constant $K \in [1, \infty)$ such that for all $x \in I$, we have

$$K^{-1} \text{dist}(x, I^c) \leq \sigma(x) \leq \beta \text{dist}(x, I^c).$$

- (ii) If $I = \mathbb{R}$, then there are constants $K \in [1, \infty)$ and $\gamma \in [0, 2]$ such that for all $x \in \mathbb{R}$, we have

$$K^{-1} \langle x \rangle^{\gamma/2} \leq \sigma(x) \leq (\beta^2 |x|^2 + K)^{1/2} \wedge (K \langle x \rangle^{\gamma/2}).$$

- (iii) If I is the half-line (a, ∞) or $(-\infty, a)$, then there are constants $K \in [1, \infty)$ and $\gamma \in [0, 2]$ such that for all $x \in I$, we have

$$K^{-1}(|x - a| \wedge |x - a|^{\gamma/2}) \leq \sigma(x) \leq (\beta |x - a| \wedge (K |x - a|^{\gamma/2})).$$

Then we have $\overline{Q}_{\text{FBSDE}}(\sigma) \geq \beta^{-2}$.

Corollary 3.2.3. Let $\sigma \in \text{Lip}(\mathbb{R}; \mathbb{R}_{\geq 0})$ and suppose there exist $\beta \in (0, \infty)$ and $k \in (0, \beta]$ such that $k|b| \leq \sigma(b) \leq \beta|b|$ for all $b \in \mathbb{R}$. Then $\overline{Q}_{\text{FBSDE}}(\sigma) \geq \beta^{-2}$.

Proof. By the previous theorem, since $\sigma(b) \geq k|b| > 0$ for any nonzero b , so we may know that for $(0, \infty)$, let $\gamma = 2, K = k^{-1} \vee 1 \vee \beta$ and then we know

$$K^{-1}|b| \leq k|b| \leq \sigma|b| \leq \beta|b| \leq \beta|b| \wedge K|b|$$

and it is similar for $(-\infty, 0)$ and we are done. $\square$

4 SPDE

5 Open Problem

One possible question is that if σ is bounded, then in term of the solution u_ϵ of the stochastic pde

$$\partial_t u_\epsilon = \frac{1}{2} \Delta u_\epsilon + \sigma(u) dW^\epsilon,$$

what will u_ϵ converges to as $\epsilon \rightarrow 0$.

5.1 Limit behavior of the parabolic equation

In this part, we consider a quasi-linear parabolic equation

$$u_t = \frac{1}{2} u u_{xx}, \quad t \in [0, \infty), x \in \mathbb{R}$$

and according to the paper by Alex.D and Cole.G, we have such conclusions about the existence and regularity of the solution of this equation. Firstly, let us define the weak solution for the equation

Definiton 5.1.1 (Weak solution). Given $T > 0$, we say that $u \in \mathcal{C}([0, T]; W_{\text{loc}}^{1, \infty}(\mathbb{R}))$ is a *weak solution* of the parabolic equation on $[0, T)$ if $u \geq 0$ and for all $\psi \in C_c^\infty([0, T) \times \mathbb{R})$,

$$\int_{\mathbb{R}} u_0 \psi(0, \cdot) + \int_{[0, T) \times \mathbb{R}} \left[u \partial_t \psi - \frac{1}{2} \partial_x(\psi u) \partial_x u \right] = 0. \quad (1)$$

With better regularity of u and $\partial_x^2 u$, we have

Definiton 5.1.2 (Strong solution). Given $\bar{T} > 0$, a weak solution of the parabolic equation on $[0, \bar{T})$ is called *strong* if for every $T \in (0, \bar{T})$ we have

$$u \in \langle x \rangle^2 L^\infty([0, T] \times \mathbb{R}) \quad \text{and} \quad \partial_x^2 u \in L^1([0, T]; L^\infty(\mathbb{R})).$$

where

$$\langle x \rangle = \sqrt{(x^2 + 1)}$$

is the Japanese braacket.

For merely weak solution, we have a lemma to secure the smoothness

Lemma 5.1.1. Let u be a weak solution of the parabolic equation on a time interval $[0, T)$. Then on the set $\{(t, x) \in (0, T) \times \mathbb{R} | u(t, x) > 0\}$, u is qualitatively smooth and satisfies the parabolic equation pointwise.

By the way, the equation admits some kind of comparsion principle

Proposition 5.1.2. Given $T > 0$, suppose $u^\pm \in \mathcal{C}_{\text{loc}}([0, T] \times \mathbb{R})$ satisfy $\partial_x^2 u^\pm \in L_t^1 L_x^\infty$ and $\partial_t u \in \langle x \rangle^2 L_t^1 L_x^\infty$ as well as

$$\partial_t u^+ \geq \frac{1}{2} u^+ \partial_x^2 u^+, \quad \partial_t u^- \leq \frac{1}{2} u^- \partial_x^2 u^-, \quad a.e.$$

Then if $u^-(0, \cdot) \leq u^+(0, \cdot)$, we have $u^- \leq u^+$.

5.1.1 Main Conclusion

Now we would like to consider for some initial condition with relatively good properties, if the limit of u at the time of infinity will exist and what it is, and a basic observation is that $u(t, \cdot)$ will approach to some constant function on $\mathbb{R}$ in general, so the we have a claim

Theorem 5.1.3 (Limit Behavior for Periodic IC). Assume $u \in \mathcal{C}_{\text{loc}}([0, \infty) \times \mathbb{R})$ is a weak solution of the parabolic equation, with a positive bounded initial condition $u_0 = u(0, \cdot) > 0$ and $\partial_x^2 u \in L_{\text{loc};t}^1 L_x^\infty$, and u_0 is periodic. Then

$$\lim_{t \rightarrow \infty} u(t, x) = C(u_0) \quad \text{for a.e. } x \in \mathbb{R},$$

where $C(u_0)$ is some constant only related to the initial condition.

Theorem 5.1.4 (Limit Behavior for Step-like IC). Assume $u \in \mathcal{C}_{\text{loc}}([0, \infty) \times \mathbb{R})$ is a weak solution of the parabolic equation, with the initial condition u_0 which satisfies that there exists a function $f \in \overline{C^\infty}^u$ with a compact support and a constant $c > 0$, such that

$$u_0 = f \text{ on some bounded interval } [a, b], \quad u_0 = \begin{cases} f(a) & \text{on } [-\infty, a] \\ f(b) & \text{on } [b, +\infty] \end{cases}$$

and $\partial_x^2 u \in L_{\text{loc};t}^1 L_x^\infty$. Then

$$\lim_{t \rightarrow \infty} u(t, x) = C(u_0) \quad \text{for a.e. } x \in \mathbb{R},$$

where $C(u_0)$ is some constant only related to the initial condition.

Lemma 5.1.5. For a weak solution u of the parabolic equation, with $\partial_x^2 u \in L_{\text{loc};t}^1 L_x^\infty$, if there exists two constant m, M such that $m \leq u_0 : u(0, \cdot) \leq M$, then $m \leq u \leq M$ on $[0, \infty) \times \mathbb{R}$.

Proof. Let M be u^+ and since u is a weak solution, so it is continuous automatically by the lemma. And apply the comparison principle pointwisely, we know $u \leq M$ for every point contained in a compact subset of $[0, \infty) \times \mathbb{R}$ and hence every point. The same argument may be applied to m . $\square$

Now let us prove the case of periodic IC.

Proof. By the lemma above, we may know that there exists $m, M > 0$ such that $m \leq u \leq M$ and hence the equation is not degenerated at every point.

Notice here $\partial_t u, \partial_x u, \partial_x^2 u$ are all defined naturally and even smooth by the smoothness of weak solution, so we may consider the energy estimation for u . Define

$$E(t) = \int_{-L}^L (\partial_x u(t, y))^2 dy$$

and then by compactness of $[-L, L]$ and DCT, we have

$$\begin{aligned}\frac{dE(t)}{dt} &= \int_{-L}^L 2\partial_{xt}u(t, y)\partial_x u(t, y)dy \\ &= 2 \left(u_t(t, x)u_x(t, x)|_{-L}^L - \int_{-L}^L u_t(t, y)\partial_{xx}u(t, y)dy \right) \\ &= - \int_{-L}^L u(t, y)(\partial_{xx}u(t, y))^2 dy \leq 0\end{aligned}$$

in both cases. So $u(t, \cdot)$ is bounded in $H^1([-L, L])$. Notice $E(t) \geq 0$ and

$$\frac{dE(t)}{dt} \leq -m \int_{-L}^L (\partial_{xx}u(t, y))^2 dy \leq 0$$

which means we may choose a subsequence u_{t_n} such that $\|\partial_x^2 u(t_n, \cdot)\|_{L^2([-L, L])} \leq 1/n$. By Morrey's inequality, $u(t_n, \cdot), \partial_x u(t_n, \cdot)$ are both compactly embedded in $C^{0,1/2}([-L, L])$ and hence we may choose a further subsequence $u(t_{n_j}, \cdot) \rightarrow u_\infty$ with $\partial_x u(t_{n_j}, \cdot) \rightarrow u'_\infty$ in $C^{0,1/2}([-L, L])$ for some u_∞ and u'_∞ and $u'_\infty = \partial_x u_\infty$ in weak sense. Therefore, $u_\infty \in W^{1,\infty}([-L, L])$.

We consider $\partial_x^2 u_\infty$ only as a distribution, then for any $\phi \in C_c^\infty([-L, L])$, we have

$$\begin{aligned}\langle \partial_t u(t_n, \cdot), \phi \rangle &= \frac{1}{2} \int_{[-L, L]} u(t_n, \cdot) \partial_x^2 u(t_n, \cdot) \phi \\ &= \frac{1}{2} \left(\int_{[-L, L]} \partial_x(u(t_n, \cdot) \partial_x u(t_n, \cdot)) - (\partial_x u(t_n, \cdot))^2 \phi \right) \\ &= \frac{1}{4} \int_{[-L, L]} \partial_x^2(u^2(t_n, \cdot)) \phi - \frac{1}{2} \int_{[-L, L]} (\partial_x u(t_n, \cdot))^2 \phi \\ &= \frac{1}{4} \int_{[-L, L]} u^2(t_n, \cdot) \partial_x^2 \phi - \frac{1}{2} \int_{[-L, L]} (\partial_x u(t_n, \cdot))^2 \phi\end{aligned}$$

since $u_\infty, \partial_x u_\infty$ are both continuous function on $[-L, L]$ and $u(t_n, \cdot) \rightarrow u_\infty, \partial_x u(t_n, \cdot) \rightarrow \partial_x u_\infty$ uniformly, we have $u^2(t_n, \cdot) \rightarrow u_\infty^2, (\partial_x u(t_n, \cdot))^2 \rightarrow (\partial_x u_\infty)^2$ uniformly. Therefore we have

$$\begin{aligned}\langle \partial_t u(t_n, \cdot), \phi \rangle &= \frac{1}{4} \int_{[-L, L]} u^2(t_n, \cdot) \partial_x^2 \phi - \frac{1}{2} \int_{[-L, L]} (\partial_x u(t_n, \cdot))^2 \phi \\ &\rightarrow \frac{1}{4} \int_{[-L, L]} u_\infty^2 \partial_x^2 \phi - \frac{1}{2} \int_{[-L, L]} (\partial_x u_\infty)^2 \phi \\ &= \frac{1}{2} \langle u_\infty \partial_x^2 u_\infty, \phi \rangle\end{aligned}$$

where $u_\infty \partial_x^2 u_\infty$ is well-defined since u_∞ is positive continuous function, which may be approached by smooth function. Notice

$$\|\partial_t u(t_n, \cdot)\|_{L^2(-L, L)} \leq \frac{1}{m} \|\partial_x^2 u(t_n, \cdot)\|_{L^2(-L, L)} \rightarrow 0$$

and hence $vv_\infty = 0$ as a distribution, which means

$$\partial_x^2(u_\infty^2) = 2(\partial_x u_\infty)^2 \in \mathcal{C}([-L, L])$$

(Check) We consequently have $\partial_x^2 u_\infty \in L^\infty([-L, L])$ by some direct computation. Now $u_\infty \partial_x^2 u_\infty$ may be interpreted by $L^\infty([-L, L])$ functions and then $\partial_x^2 u_\infty = 0$ by u_∞ positive, so u_∞ is some constant between m and M when u_0 is periodic, and the constant is exactly c as we defined in another case. We denote this constant as u_∞ uniformly for the following.

Now we may consider the truncation solution U_T which is the solution of the parabolic equation with the initial condition $u(T, \cdot)$, by the comparison principle, we know U_T will be bounded in $[u_\infty - \epsilon, u_\infty + \epsilon]$ if $|u(T, \cdot) - u_\infty| < \epsilon$. To sum up, let T to be t_n we chose before and then we will know $u(t + t_n, x) = U_T(t, x) \in [u_\infty - \epsilon_n, u_\infty + \epsilon_n]$ for any $t \geq 0$ and $\epsilon_n \rightarrow 0$ with $n \rightarrow \infty$, so we obtained that

$$\lim_{t \rightarrow \infty} u(t, \cdot) = u_\infty$$

in $\mathcal{C}(\mathbb{R})$, and in particular, for any $x \in \mathbb{R}$,

$$\lim_{t \rightarrow \infty} u(t, x) = u_\infty.$$

□

Lemma 5.1.6. For a weak solution u , with positive lower bound and upper bound, of the parabolic equation and smooth initial condition u_0 , with $\partial_x^2 u \in L_{\text{loc};t}^1 L_x^\infty$, if there are limits

$$\lim_{x \rightarrow -\infty} u_0(x), \quad \lim_{x \rightarrow +\infty} u_0(x)$$

exist, then $\lim_{x \rightarrow \infty} \partial_x u(t, x) = 0$.

Proof. We want to show that

$$\lim_{x \rightarrow -\infty} u(x, t) = \lim_{x \rightarrow -\infty} u_0(x), \quad \lim_{x \rightarrow +\infty} u(x, t) = \lim_{x \rightarrow +\infty} u_0(x).$$

By applying Gaussian estimation for $v = u - \lim_{x \rightarrow -\infty} u_0(x)$, which satisfies a uniformly elliptic equation

$$v_t = \frac{u}{2} \partial_x^2 v$$

and we have for any $t > 0$,

$$|v(t, x)| \leq \int \frac{C}{t^{n/2}} \exp\left(-\frac{C'|x-y|^2}{t}\right) v_0(y) dy \rightarrow 0, \quad \text{if } x \rightarrow -\infty$$

for some positive constants C, C' and $v_0 = v(0, \cdot)$. Therefore, we know for any $t > 0$, we have

$$\lim_{x \rightarrow -\infty} u(x, t) = \lim_{x \rightarrow -\infty} v(x, t) + \lim_{x \rightarrow -\infty} u_0(x) = \lim_{x \rightarrow -\infty} u_0(x)$$

and the other equality can be shown in an analogous argument.

Now since the initial condition is smooth with positive lower and upper bound, we may use the Schauder estimation for u and hence for any $0 < \alpha < 1, T > 0$, we have on $[0, T] \times \mathbb{R}$

$$\|u\|_{C_b^{2+\alpha, 1+\alpha/2}} \leq C(\|u_0\|_{C_b^{2+\alpha}}, m, M, T)$$

and hence $\sup_{t \in [0, T]} \|\partial_x^2 u\|_{L^\infty} < \infty$. Then for any $t \leq T$ we know $\lim_{x \rightarrow \infty} \partial_x u = 0$ since

$\partial_x u(t, \cdot)$ is Lipschitz. □

The case for bump IC is a little different, here is the proof.

Proof. We use the same notation of m, M as the proof in periodic case. Assume $u_0 \in C^\infty$ at first.

For the energy estimation, we consider

$$E_L(t) = \int_{-L}^L (\partial_x u(t, y))^2 dy$$

and $E(t) := E_\infty(t)$, then we may know that

$$\begin{aligned} \frac{dE_L(t)}{dt} &= 2u_t(t, x)\partial_x u(t, x)|_{-L}^L - \int_{-L}^L u(t, y)(\partial_x^2 u(t, y))^2 dy \\ &= u(t, x)\partial_x(t, x)\partial_x^2 u(t, x)|_{-L}^L - \int_{-L}^L u(t, y)(\partial_x^2 u(t, y))^2 dy \end{aligned}$$

and since $\|\partial_x^2 u(t, \cdot)\|_{L^\infty}$ (should be guaranteed by some other estimation) is bound on $[0, T]$ for any T . By the lemma above, we know

$$\lim_{L \rightarrow \infty} \frac{dE_L(t)}{dt} = - \int u(t, y)(\partial_x^2 u(t, y))^2 dy$$

for almost every t . On the other hand, notice

$$\frac{dE_L(t)}{dt} \leq u(t, x)\partial_x(t, x)\partial_x^2 u(t, x)|_{-L}^L$$

so for $T > 0$, we may choose $L_{n;T}$ such that $u(t, x)\partial_x(t, x)\partial_x^2 u(t, x)|_{-L}^L \leq 1/n$ if $L > L_{n;T}$ and hence $\frac{dE_{L_{n;T}}(t)}{dt} \leq 1/n$ for any $t \in [0, T]$. Then

$$E_{L_{n;T}}(t) \leq E_{L_{n;T}}(0) + T/n \leq E(0) + T/n$$

for any $t \in [0, T]$, and let $n \rightarrow \infty$, we may know $E(t) \leq E(0)$ for any $t > 0$. Moreover, $E(t)$ is decreasing. For any $\delta > 0$, if then are T_δ such that for any $t > T_\delta$, $\|\partial_x^2 u(t, \cdot)\|_{L^2} > \delta$, then we know for $L > L_{n;MT_\delta}$

$$\frac{dE_L(t)}{dt} = 1/n - I_L(t)$$

where $I_L(t) := \int_{-L}^L u(t, y)(\partial_x^2 u(t, y))^2 dy$ and hence

$$E_L(MT_\delta) \leq E_L(T_\delta) - \left(\int_{T_\delta}^{MT_\delta} I_L(t) dt - (M-1)T_\delta/n \right)$$

so we may choose M sufficiently large such that $\frac{(M-1)}{2}T_\delta m\delta > 2E(0)$, and choose n large enough such that $1/n < m\delta/4$ and L large enough such that

$$\int_{-L}^L \int_{T_\delta}^{MT_\delta} u(t, y)(\partial_x^2 u(t, y))^2 dy dt = \int_{T_\delta}^{MT_\delta} I_L(t) dt > 3m(M-1)T_\delta\delta/4$$

since $\int_{T_\delta}^{MT_\delta} u(t, y)(\partial_x^2 u(t, y))^2 dy dt > m(M-1)T_\delta\delta$ by Fubini and hence L can be chosen by MCT. Then we know that $E_{L_n; MT_\delta}(MT_\delta) \leq E(0) - 2E(0) < 0$, which will be a contradiction, so we know we may choose a subsequence $u(t_n, \cdot)$ such that

$$\|\partial_x^2 u(t_n, \cdot)\|_{L^2} \rightarrow 0$$

Therefore, we may use the analog argument for periodic case to show that u_∞ will be a constant.

Now let show that the conclusion is true for any $f \in \overline{C^\infty}^u$ with compact support. For any $\epsilon > 0$, we may consider that ϕ_ϵ to be some smooth function with compact support and $\|\phi_\epsilon - f\| < \epsilon$, then we know

$$\phi_\epsilon - \epsilon + c < u_0 = f + c < \phi_\epsilon + \epsilon + c.$$

Consider the weak solutions $\phi_\epsilon^-, \phi_\epsilon^+$ given by $\phi_\epsilon - \epsilon + c, \phi_\epsilon + \epsilon + c$. By the comparsion principle,

$$\phi_{\epsilon, \infty}^- < \phi_{\epsilon, \infty}^+, \quad \phi_{\epsilon, \infty}^- > \phi_{\epsilon, \infty}^+ - 2\epsilon, \quad \phi_{\epsilon, \infty}^- \leq \liminf_{t \rightarrow \infty} u \leq \limsup_{t \rightarrow \infty} u \leq \phi_{\epsilon, \infty}^+$$

and it is easy to check that

$$u_\infty = \limsup_{\epsilon \rightarrow 0} \phi_{\epsilon, \infty}^-$$

□

5.1.2 The value of Weak Solution at Infinity Time

Theorem 5.1.7 (Limit Value for Periodic IC). The limit

$$u_\infty = \exp \left(\frac{1}{L} \int_0^L \ln u_0(y) dy \right)$$

in the periodic initial condition case, where L is a period length of u_0 .

Proof. We consider the value

$$\Phi(t) = \int_0^L \ln(u(t, y)) dy$$

and then

$$\frac{d\Phi}{dt} = \int_0^L u^{-1}(t, y) u_t(t, y) dy = \frac{1}{2} u_x(t, \cdot)|_0^L = 0$$

which means $\Phi(t) = \Phi(0) = \int_0^L \ln u_0(y) dy$ for any t , however, by DCT we know

$$\lim_{t \rightarrow \infty} \Phi(t) = L \ln u_\infty$$

so we have are done. □

Theorem 5.1.8 (Limit Value for Bump IC). In the case of step-like initial condition, if we have same sides limits, i.e. $f(a) = f(b) =: c$, then limit behavior is consistent even f is

merely continuous, and the limit value $u_\infty = c$.

Proof. Assume f is nonnegative, we consider the periodic solution $u^{(L)}$ defined by the following initial condition,

$$u_0^{(L)} = u_0 \text{ on } [-L, L] \text{ with period length } 2L,$$

and then $u_0^{(L)} \geq u_0$ for any L large enough. We know

$$u_\infty^{(L)} = \exp\left(\frac{1}{2L} \int_{-L}^L \ln u_0^{(L)}(y) dy\right) \rightarrow c, \text{ with } L \rightarrow \infty$$

and notice that in this case, by comparison principle $u_\infty^{(L)} \geq \lim_{t \rightarrow \infty} \sup_x u(t, x) \geq c$ and notice $u \geq c$, so $u_\infty = c$. We then use the analogous argument for f nonpositive to show $u_\infty = c$ as well.

For general f , we may consider $h = (f - c)^+ + c$ and $g = (f - c)^- + c$, then we know $h_\infty = g_\infty = c$ with

$$h_\infty \geq \lim_{t \rightarrow \infty} \sup_x u(t, x) \geq \lim_{t \rightarrow \infty} \inf_x u(t, x) \geq g_\infty$$

□

$$|u_x(t, \cdot)| \sim \epsilon/t$$

$$\|u_x(t, \cdot)\|_{L^2(\mathbb{T}_L)} \leq \exp\left(-\frac{1}{2}m\left(\frac{2\pi}{L}\right)^2 t\right) \|u_x(0, \cdot)\|_{L^2(\mathbb{T}_L)}.$$

$$\|u_x(t, \cdot)\|_{L^\infty(\mathbb{T}_L)} \leq \exp\left(-\frac{1}{2}m\left(\frac{2\pi}{L}\right)^2 t\right) \|u_x(0, \cdot)\|_{L^2(\mathbb{T}_L)} \sim \exp(-Ct/\epsilon^2) \cdot \frac{1}{\epsilon} \leq \frac{\epsilon^2}{t} \frac{1}{\epsilon}$$

$$\|u_x\|_{L^2}^3 \leq C\|u_x\|_{L^1}^2 \|u_{xx}\|_{L^2}$$

$$\frac{d}{dt}\|u_x\|_{L^2} \leq -C'\|u_{xx}\|_{L^2}$$

$$u = u_\infty + v, v \rightarrow 0 \text{ uniformly}$$

$$v_t = \frac{1}{2}u_\infty v_{xx} + \frac{1}{2}v v_{xx} := f$$

$$u = c\Delta u + f$$

$$u(t, \cdot) = G_{ct}u_0 + \int_0^t G_{c(t-s)}f(s, \cdot)ds$$

$$u_x(t, \cdot) = G_{ct}u_x(0, \cdot) + \int_0^t G_c(t-s)f_x(s, \cdot)ds$$

where

$$f_x = \frac{1}{2}(v_x v_{xx} + v v_{xxx})$$

Proposition 5.1.9.

Poincaré inequality on the periodic interval.

Let $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$. If $f \in H^1(\mathbb{T}_L)$ and

$$\int_{\mathbb{T}_L} f(x) dx = 0,$$

then

$$\int_{\mathbb{T}_L} f(x)^2 dx \leq \left(\frac{L}{2\pi}\right)^2 \int_{\mathbb{T}_L} f_x(x)^2 dx.$$

Equivalently,

$$\|f\|_{L^2(\mathbb{T}_L)} \leq \frac{L}{2\pi} \|f_x\|_{L^2(\mathbb{T}_L)}.$$

Proof. Since f is L -periodic, we may write its Fourier series as

$$f(x) = \sum_{k \in \mathbb{Z}} \hat{f}_k e^{2\pi i k x / L},$$

where

$$\hat{f}_k = \frac{1}{L} \int_0^L f(x) e^{-2\pi i k x / L} dx.$$

The zero-average condition implies

$$\hat{f}_0 = \frac{1}{L} \int_0^L f(x) dx = 0.$$

Hence

$$f(x) = \sum_{k \in \mathbb{Z} \setminus \{0\}} \hat{f}_k e^{2\pi i k x / L}.$$

Differentiating term by term in the weak sense, we get

$$f_x(x) = \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{2\pi i k}{L} \hat{f}_k e^{2\pi i k x / L}.$$

By Parseval's identity,

$$\int_0^L |f(x)|^2 dx = L \sum_{k \in \mathbb{Z}} |\hat{f}_k|^2.$$

Since $\hat{f}_0 = 0$, this becomes

$$\int_0^L |f(x)|^2 dx = L \sum_{k \neq 0} |\hat{f}_k|^2.$$

Similarly,

$$\int_0^L |f_x(x)|^2 dx = L \sum_{k \neq 0} \left| \frac{2\pi k}{L} \right|^2 |\hat{f}_k|^2.$$

For $k \neq 0$, we have $k^2 \geq 1$. Therefore

$$\left| \frac{2\pi k}{L} \right|^2 \geq \left(\frac{2\pi}{L} \right)^2.$$

It follows that

$$\begin{aligned}
\int_0^L |f_x(x)|^2 dx &= L \sum_{k \neq 0} \left| \frac{2\pi k}{L} \right|^2 |\widehat{f}_k|^2 \\
&\geq L \sum_{k \neq 0} \left(\frac{2\pi}{L} \right)^2 |\widehat{f}_k|^2 \\
&= \left(\frac{2\pi}{L} \right)^2 L \sum_{k \neq 0} |\widehat{f}_k|^2 \\
&= \left(\frac{2\pi}{L} \right)^2 \int_0^L |f(x)|^2 dx.
\end{aligned}$$

Thus

$$\int_0^L |f(x)|^2 dx \leq \left(\frac{L}{2\pi} \right)^2 \int_0^L |f_x(x)|^2 dx.$$

This proves the desired inequality. $\square$

Proposition 5.1.10 (Energy Estimation).

Let

$$p(t, x) := u_x(t, x).$$

Since

$$u_t = \frac{1}{2} u u_{xx},$$

we differentiate in x and obtain

$$p_t = \partial_x u_t = \frac{1}{2} \partial_x (u u_{xx}) = \frac{1}{2} \partial_x (u p_x).$$

Therefore

$$\boxed{p_t = \frac{1}{2} (u p_x)_x.}$$

Assume that u is L -periodic in x , and let

$$\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}.$$

Since $p = u_x$, we have

$$\int_{\mathbb{T}_L} p(t, x) dx = \int_0^L u_x(t, x) dx = u(t, L) - u(t, 0) = 0.$$

Define the energy

$$E(t) := \int_{\mathbb{T}_L} p(t, x)^2 dx.$$

Then

$$\frac{d}{dt} E(t) = \frac{d}{dt} \int_{\mathbb{T}_L} p^2 dx = 2 \int_{\mathbb{T}_L} p p_t dx.$$

Using

$$p_t = \frac{1}{2} (u p_x)_x,$$

we get

$$\frac{d}{dt}E(t) = 2 \int_{\mathbb{T}_L} p \cdot \frac{1}{2}(up_x)_x dx = \int_{\mathbb{T}_L} p(up_x)_x dx.$$

By integration by parts and periodicity,

$$\int_{\mathbb{T}_L} p(up_x)_x dx = - \int_{\mathbb{T}_L} p_x up_x dx.$$

Hence

$$E'(t) = - \int_{\mathbb{T}_L} u(t, x) p_x(t, x)^2 dx.$$

If

$$u(t, x) \geq m > 0,$$

then

$$E'(t) \leq -m \int_{\mathbb{T}_L} p_x(t, x)^2 dx.$$

By the Poincaré inequality, since $\int_{\mathbb{T}_L} p dx = 0$,

$$\int_{\mathbb{T}_L} p_x^2 dx \geq \left(\frac{2\pi}{L}\right)^2 \int_{\mathbb{T}_L} p^2 dx.$$

Therefore

$$E'(t) \leq -m \left(\frac{2\pi}{L}\right)^2 E(t).$$

By Grönwall's inequality,

$$E(t) \leq \exp\left(-m \left(\frac{2\pi}{L}\right)^2 t\right) E(0).$$

That is,

$$\int_{\mathbb{T}_L} u_x(t, x)^2 dx \leq \exp\left(-m \left(\frac{2\pi}{L}\right)^2 t\right) \int_{\mathbb{T}_L} u_x(0, x)^2 dx.$$

Equivalently,

$$\|u_x(t, \cdot)\|_{L^2(\mathbb{T}_L)} \leq \exp\left(-\frac{1}{2}m \left(\frac{2\pi}{L}\right)^2 t\right) \|u_x(0, \cdot)\|_{L^2(\mathbb{T}_L)}.$$

A Fundamentals

A.1 Wiener Integral

Let T be a set and $X := \{X(t)\}_{t \in T}$ a T -indexed stochastic process. We recall that X is a Gaussian random field (process when $T \subset \mathbb{R}$) if $(X_{t_1}, \dots, X_{t_m})$ is a Gaussian random vector for all $t_1, \dots, t_m \in T$.

Definiton A.1.1. Let $\mathcal{B}(\mathbb{R}^m)$ denote the collection of all Borel-measurable subsets of $\mathbb{R}^m$ that have finite Lebesgue measure. White noise on $\mathbb{R}^m$ is a mean-zero set-indexed Gaussian random field $\xi(A)_{A \in \mathcal{B}(\mathbb{R}^m)}$ with covariance function

$$E[\xi(A_1)\xi(A_2)] := |A_1 \cap A_2| \quad \text{for all } A_1, A_2 \in \mathcal{L}(\mathbb{R}^m),$$

where $|\cdot|$ denotes the Lebesgue measure on $\mathbb{R}^m$ for every m .

Remark. Here is a general proof from Ash and Gardner.

Proof. For any complex numbers $c_1, \dots, c_n$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}^m)$, we have

$$\sum_{1 \leq i, j \leq n} c_i \bar{c}_j |A_i \cap A_j| = \sum_{1 \leq i, j \leq n} c_i \bar{c}_j \int \chi(A_i) \chi(A_j) = \int \left| \sum_{i=1}^n c_i \chi(A_i) \right|^2 \geq 0$$

and we are done. $\square$

Lemma A.1.1. $\{\xi(A)\}_{A \in \mathcal{B}(\mathbb{R}^m)}$ is an $L^2(\Omega)$ -valued countably-additive measure on $\mathcal{B}(\mathbb{R}^m)$.

Definiton A.1.2 (Wiener Integral). For $h \in \mathcal{S}(\mathbb{R}^m)$ the set of the simple functions, define

$$\int h d\xi := \sum_{i=1}^n c_i \xi(A_i)$$

where $h(x) = \sum_{i=1}^n c_i \chi(A_i)$ for some $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R}^m)$ and $c_1, \dots, c_n \in \mathbb{R}$.

For $f \in L^2(\mathbb{R}^m)$, define

$$\int f d\xi = \lim_{n \rightarrow \infty} \int f_i d\xi$$

in $L^2(\Omega)$ for some $f_i \in \mathcal{S}(\mathbb{R}^m)$ and $f_i \rightarrow f$ in L^2 .

Proof. $\square$

Proposition A.1.2. The $L^2(\mathbb{R}^m)$ -indexed Gaussian random field $\{\int h d\xi\}_{h \in L^2(\mathbb{R}^m)}$ has the following properties:

1. $\int (a_1 h_1 + a_2 h_2) d\xi = a_1 \int h_1 d\xi + a_2 \int h_2 d\xi$ a.s. for all $a_1, a_2 \in \mathbb{R}$ and $h_1, h_2 \in L^2(\mathbb{R}^m)$.
2. For all $h, h_1, h_2 \in L^2(\mathbb{R}^m)$,

$$\mathbb{E} \int h d\xi = 0, \mathbb{E} \left(\int h_1 d\xi \int h_2 d\xi \right) = \int h_1 h_2$$

Remark. We may define

$$\int_A h d\xi = \int h \chi(A) d\xi$$

and it is easy to prove the *Wiener isometry* by the properties above.

Definiton A.1.3. For every $f \in L^2(\mathbb{R}^m)$ and $x \in \mathbb{R}^m$, define

$$(f * \xi)(x) := \int f(x - y)\xi(dy)$$

Proposition A.1.3. For every $f \in L^2(\mathbb{R}^m)$, there exists a version of $f * \xi$ that is measurable jointly in (ω, x) . Moreover, we have the following Young-type inequality: For every finite Borel measure μ on $\mathbb{R}^m$,

$$\mathbb{E} \left(\left| \int |(f * \xi)(x)| \mu(dx) \right|^2 \right) \leq \mu(\mathbb{R}^m)^2 \|f\|_{L^2(\mathbb{R}^m)}^2$$

Proof. □

A.2 Useful Approximations

Proposition A.2.1 (Bihari-LaSalle Inequality).

Let u and f be non-negative continuous functions defined on the half-infinite ray $[0, \infty)$ and let w be a continuous non-decreasing function defined on $[0, \infty)$ and $w(u) > 0$ on $(0, \infty)$. For u we have

$$u(t) \leq \alpha + \int_0^t f(s)w(u(s))ds, \quad t \in [0, \infty),$$

where α is a non-negative constant, then

$$u(t) \leq G^{-1} \left(G(\alpha) + \int_0^t f(s)ds \right), \quad t \in [0, T]$$

where the function G is defined by

$$G(x) = \int_{x_0}^x \frac{dy}{w(y)}, \quad x \geq 0, x_0 \geq 0,$$

and G^{-1} is the inverse function of G and T is chosen so that

$$G(\alpha) + \int_0^t f(s)ds \in \text{Dom}(G^{-1}), \quad \forall t \in [0, T]$$

Proof. Notice $G(x) = \int_{x_0}^x \frac{dy}{w(y)}$ is increasing and

$$\text{Dom}(G^{-1}) = \left[G(0) - \int_0^{x_0} \frac{dy}{w(y)}, G(0) + \int_0^\infty \frac{dy}{w(y)} \right]$$

then since $u \geq 0$, then we have

$$G(u(t)) \leq G \left(\alpha + \int_0^t f(s)\omega(u(s))ds \right) := H(t)$$

and we have

$$H'(t) = \frac{f(t)\omega(u(t))}{\omega(\alpha + \int_0^t f(s)\omega(u(s))ds)} \leq f(t)$$

so we have

$$H(t) \leq G(\alpha) + \int_0^t f(s)ds.$$

Therefore, we may have the requested inequality for all $0 \leq t \leq T$ where

$$T := \sup_t \left\{ \int_0^t f(s)ds \leq G(0) - G(\alpha) + \int_0^\infty \frac{dy}{\omega(y)} \right\}$$

□

A.3 Matrix Analysis

Theorem A.3.1 (Spectral Theorem for Real Symmetric Matrices).

Let $A \in \mathbb{R}^{n \times n}$ be a real symmetric matrix, that is, $A^T = A$. Then:

1. All eigenvalues of A are real.
2. Eigenspaces corresponding to distinct eigenvalues are orthogonal.
3. The space $\mathbb{R}^n$ decomposes as an orthogonal direct sum of eigenspaces:

$$\mathbb{R}^n = \bigoplus_{\lambda \in \sigma(A)} \ker(A - \lambda I).$$

4. Equivalently, there exists an orthogonal matrix Q and a real diagonal matrix Λ such that

$$A = Q\Lambda Q^T.$$

A.4 Kolmogorov's Theorems

Definiton A.4.1. For any $w \in \mathbb{R}^m$, then

$$\rho(w) := \sum_{j=1}^m |w_j|^{\alpha_j}$$

where $0 < \alpha_1, \dots, \alpha_m \leq 1$ are fixed.

For a bounded and closed subset T of $\mathbb{R}^m$, define

$$\Lambda_T(a) := \int_T dt \int_T ds [\rho(t-s)]^{-a} \quad \text{for all } a > 0.$$

Remark. $\Lambda_T(a) < \infty$ if and only if $a < H := \sum_{j=1}^m \alpha_j^{-1}$.

Theorem A.4.1 (Kolmogorov's Continuity Theorem).

Let $\{X_t\}_{t \in T}$ be a stochastic process, where $T \subset \mathbb{R}^m$ is measurable and bounded, and suppose that there exists finite real numbers $C > 0$ and $k > H$, such that

$$\|X_t - X_s\|_k \leq C\rho(t-s) \quad \text{for any } s, t \in T$$

Then X has a modification $\bar{X}$ that is Holder continuous a.s. Moreover, $\bar{X}$ satisfies the following: For all $q \in (0, 1 - (H/k))$ and $\delta \in (q, 1 - (H/k))$ there exists a finite constant D ,

that does not depend on that numerical value of k , such that

$$\mathbb{E} \left(\sup_{s, t \in T, s \neq t} \left| \frac{\overline{X}_t - \overline{X}_s}{[\rho(t-s)]^q} \right|^k \right) \leq (\delta - q)^{-1} \left(\frac{DC}{\delta} \right)^k k^{2H} \Lambda_T(2H - k + k\delta)$$